TITLE: Medicinal Mushrooms for Cognition and Mood: A review of human clinical trials

Keywords: medicinal mushrooms, cognition, mood disorders, *Hericium erinaceus*, *Ganoderma lucidum*

ABSTRACT

Medicinal mushrooms have been used since ancient times to treat disease and promote general health and well-being. Various health-promoting properties have since been confirmed with studies demonstrating immunomodulatory, anti-cancer, anti-viral, anti-bacterial, anti-inflammatory, neuroprotective, and cardioprotective activity among other properties. An additional interest in the use of mushroom nutraceuticals for mental health and cognition enhancement has also developed with various dietary supplement companies advertising their mushroom products with claims of improved memory, focus, and mood. However, research is still in its early stages with few human clinical trials to support these claims. The purpose of this review is to compile and examine the evidence for beneficial effects on the mind by medicinal mushrooms, specifically regarding the domains of cognition and mood, by reviewing human clinical trials. This review reveals trends of improvement in well-being by *G. lucidum* in the context of taxing disease states, improvement in cognitive function *H. erinaceus* in the context of cognitive decline, and general improvement in depression and anxiety by *H. erinaceus*. Potential mechanisms and limitations due to research design are also discussed. This review provides further clarity on the validity of health claims made by supplement companies and calls for future research in these areas.

INTRODUCTION

Medicinal mushrooms have been used to promote health and to treat a variety of ailments for thousands of years. Documentation of therapeutic mushroom use is present in many cultures throughout history, including the ancient Greeks, Chinese, and first peoples of North America.[1] These mushrooms were valued for their pharmaceutical properties rather than nutritional value alone. Certain mushrooms are known as functional foods, foods which may promote optimal health and reduce disease risk. They can be found in supplement form as liquid or powdered extracts known as nutraceuticals. Mycotherapy describes the use of these mushroom extracts to treat illness or support general health. A growing interest has developed in the potential medical applications of these nutraceuticals with modern research identifying and testing the activity of various bioactive compounds from these mushrooms. *In vitro* studies have shown immunomodulatory,[2,3] anticancer,[4–6] carioprotective,[7] neuroprotective,[8–10] and other health-promoting activity[11] from a variety of mushroom species.

Ophiocordyceps sinensis, previously known as *Cordyceps sinensis*, is a well-known medicinal mushroom in traditional Chinese medicine where it is called *DongChongXiaCao*. It is also nicknamed caterpillar fungus due to its parasitic relationship with caterpillars. Within traditional medicine, this mushroom is thought to benefit the kidneys, lungs, liver, circulation, cardiovascular system, and to reduce fatigue.[12] Modern research indicate antitumor, immunomodulatory, antioxidant, anti fatigue, hepatoprotective, and renal protective activity of *Cordyceps*,[13] though there is a need for more research in humans. *Cordyceps militaris* is a commonly used substitute for *O. sinensis* due to the rarity and costly cultivation process of *O. sinensis*. Additionally, a key bioactive compound of *O. sinensis*, cordycepin, is also found in *C. militaris* and in greater concentrations, making it a suitable alternative.[13]

Trametes versicolor was traditionally used for general health promotion, increasing strength and vitality, and promoting longevity.[14] It was previously known as *Coriolus versicolor* and is known as turkey tail in North America, *YunZhi* in China, and *Kawaratake* in Japan. Evidence for its immunomodulatory, antimicrobial, antiviral, and anticancer properties has contributed to its approved and routine clinical use in China and Japan as a complementary cancer therapy.[11,14,15] The health effects are largely attributed to the bioactive polysaccharopeptides (PSPs) from two different strains of the mushroom. The COV-1 strain produces PSP and CM101 produces polysaccharide krestin (PSK).[14]

Ganoderma lucidum, called *LingZhi* in China and *Reishi* in Japan, is thought to promote longevity according to traditional medicine. Documentation of traditional Chinese medicine practices note beneficial effects regarding cardiac function, memory, antiaging, replenishing *Qi*, easing the mind, insomnia, and relieving cough, asthma, and shortness of breath.[16] There are hundreds of bioactive compounds found in the spores, mycelia, and fruiting bodies of *G. lucidum*. These include triterpenoids, polysaccharides, nucleotides, steroids, and peptides among other compounds.[17] Properties of *G. lucidum* demonstrated by preclinical studies include antiaging, immunomodulatory, anticancer, cardioprotective, anti-inflammatory, antiviral, antifungal, and antibacterial activity.[11,18–20]

Hericium erinaceus also has a history of use in traditional Chinese medicine and is known for its bioactive hericenones, erinacines, and polysaccharides. Alternative names for this mushroom include lion's mane mushroom and monkey head mushroom, named after its voluminous, fur-like appearance. In addition to its antioxidant, antibacterial, and anticancer potential,[21,22] *H. erinaceus* has been shown to stimulate nerve-growth factor (NGF), promote oligodendrocyte maturation, and display neuroprotective activity in animal models.[23–25] *In vivo* studies have also shown improvements and protective activity in animal models of ischemic stroke, Parkinson's disease, Alzheimer's disease, and depression.[23]

A growing interest in both the therapeutic and everyday use of medicinal mushrooms has developed with the excitement surrounding the implications of recent research and the increasing accessibility of mushroom nutraceuticals. Common health claims from dietary supplement companies, such as Swanson, Fungi Perfecti, MUD\WTR, Om Mushrooms, and Fresh Cap, include immune support from *T. versicolor* [26–29]; energy, endurance, and cardiovascular benefits from *Cordyceps* [30–32]; stress relief, immune support, and mood support from *G. lucidum* [33–36]; and improved memory, focus, clarity, mood balance, and nervous system health from *H. erinaceus.*[37–40] Other claims such as increased mental clarity,[41] cognitive functioning,[42] and mental sharpness[43] are advertised for various blends of mushroom supplements.

However, with few human clinical trials measuring the cognitive impact of medicinal mushrooms, there is limited evidence supporting these claims. While evidence from *in vitro* and animal studies is valuable, the applicability in humans does not directly translate. This is why evidence from human studies is so vital for the evaluation of the efficacy of medicinal mushrooms for different applications and contexts.

While the literature for the use of medicinal mushrooms in an immunological context is more developed,[6,44] no reviews on the cognitive effects of mushroom supplementation in humans were found upon a search of the literature. This article reviews the human clinical trials measuring cognition-related outcomes to provide clarity on the current evidence.

METHODS

A search was conducted through the PubMed database. Only available, full-text, human clinical trials on the selected mushrooms which also measured at least one cognition-related outcome were included. The following mushrooms and their common names were searched: *Hericium erinaceus*, lion's mane, and monkey head mushroom; *Ganoderma lucidum*, reishi, and lingzhi; *Ophiocordyceps sinensis, Cordyceps sinensis, Cordyceps militaris*, and caterpillar fungus; *Inonotus obliquus*, and chaga; *Trametes versicolor, Coriolus versicolor, Polyporus versicolor*, and turkey tail. The following terms were used to search for cognition-related outcome measures: cognit*, mind, mental, memory, attention, focus, executive function, brain, mood, anxiety, depression, dementia, Alzheimer's, psych*, and well-being. Additional available studies referenced in other articles that met the inclusion criteria were also included.

RESULTS

The search yielded 11 studies on 4 mushrooms species. One study was found for *Cordyceps militaris,* one for *Coriolus versicolor,* 4 for *Ganoderma lucidum,* and 5 for *Hericium erinaceus*. Population, sample size, duration, dosage, and cognitive outcomes of the studies are outlined in table 1.

Cordyceps militaris

No evidence was found supporting the efficacy of *C. militaris* for improving insomnia or depression in patients with depression taking duloxetine.[45] Depression, evaluated by the 17-item Hamilton Depression Scale (HAMD-17), reduced over time in both groups, but this change was attributed to duloxetine and the difference was nonsignificant from baseline and between groups. Results from the Athens Insomnia Scale (AIS), a questionnaire assessing sleep difficulty, reflected a significantly lesser improvement in insomnia in the *C. militaris* group compared to the placebo group. Additionally, "well-being," measured as a subscale on the AIS, saw no significant change within or between groups. Furthermore, the AIS subscale of "functional capacity during the day" revealed a significantly greater improvement favoring the placebo group by the end of the intervention.

Table 1. Summary of human clinical trials

Mushroom	Citation	Study Population	Sample Size[a]	Duration of Supplementation	Dosage (per day)	Cognitive Outcomes
Cordyceps militaris	Zhou et al. (2021)[45]	Adults with major depressive disorder with insomia	Treatment (*C. militaris* + duloxetine): 28 Control (placebo + duloxetine): 31	6 weeks	3 g	No significant change nor difference between groups in "well-being" related to insomnia (measured by AIS). Significant improvement in "functioning capacity during the day" related to insomnia from baseline in both groups, but significantly greater improvement in placebo group (measured by AIS). Significantly greater improvement in insomnia (measured by AIS) in placebo group. No significantly greater reduction in depression (measured by HAMD-17) in treatment group.
Coriolus versicolor (Trametes versicolor)	Scuto et al. (2019)[46]	Adults with Meniere's disease	Treatment: 22 Control: 18	8 weeks	3 g biomass	Notable improvement in total mood disturbance, including "anger," "confusion," "depression," and "tension" (measured by POMS) in treatment group but not in control group. Significant improvement in tinnitus severity (measured by THI) in treatment group compared to control group, and significant improvements in frequency range, average loss in dB, and intellection threshold related to auditory function from baseline in the treatment group, but not control group.

Ganoderma lucidum	Zhao et al. (2012)[47]	Adult females with breast cancer	Treatment: 25 Control: 23	4 weeks	3 g spore powder	Significant improvement in "emotional well-being" and "emotional functioning" (measured by FACT-F and EORTC QLQ-C30 respectively) in treatment group only, and significant difference between groups favoring the treatment group. Significant improvement in anxiety and depression (measured by HADS) in treatment group only, and significant difference between groups favoring the treatment group. No significant change in "social/family well-being" or "social functioning" (measured by FACT-F and EORTC QLQ-30 respectively) in either group, nor significant difference between groups. Significant improvement in "cognitive functioning" (measured by EORTC QLQ-C30) in treatment group only and significant difference between groups favoring the treatment group.
	Wang et al. (2018)[48]	Adults with Alzheimer's disease	Treatment: 21 Control: 21	6 weeks	3 g spore powder	No significant difference between groups in cognition, dementia-related neuropsychiatric symptoms, and quality of life (measured by the ADAS-Cog, NPI, and WHOQOL-BREF respectively).
	Liu et al. (2020)[49]	Adults with non-small cell lung cancer undergoing chemotherapy	Treatment: 61 Control: 21	6 weeks	0.45 g spore powder[b]	Greater improvements in "social/family well-being," "emotional well-being," and "functional well-being" contributing to a higher quality of life (measured by FACT-G) in treatment group, but no significant difference between groups.

	Pazzi et al. (2020)[50]	Adult females with fibromyalgia	Treatment: 26 Control: 24	6 weeks	6 g fruiting body powder	Significant improvements in happiness, satisfaction with life, and depression (measured by SHS, SWLS, and GDS respectively) from baseline in treatment group but not placebo group with a significant difference between groups favoring the treatment group in happiness. Significant improvement in "social functioning," "emotional role" of health and "mental health" (measured by the SF-12) from baseline in treatment group and an improvement in only "mental health" in placebo, but no significant difference between groups.
Hericium erinaceus	Mori et al. (2009)[51]	Japanese adults ages 50 to 80 years old with mild cognitive impairment	Treatment: 14 Control: 15	16 weeks	1 g fruiting body powder	Significant improvement in cognitive function (measured by modified HDS-R) from baseline in treatment group and compared to placebo group.
	Nagano et al. (2010)[52]	Adult females	Treatment: 12 Control: 14	4 weeks	2 g fruiting body powder	Significant improvement in depressive symptomatology (measured by CES-D) from baseline in treatment group, but no significant difference between group outcomes. Significant improvements in “incentive” and “concentration” subscores relevant to depression (measured by ICI) from baseline in treatment group with improvement in "incentrive" significantly greater compared to placebo. Significant improvement in subscore of “irritating” (measured by ICI) from baseline in the treatment group, but no significant difference between groups. Significant improvement in “anxious” subscore (measured by ICI) from baseline in the treatment group, but no significant difference between groups.

	Saitsu et al. (2019)[53]	Healthy adults over 50 years old	Treatment: 16 Control: 15	12 weeks	3.2 g fruiting body powder	Significant improvement in cognitive function (measured by MMSE) from baseline in only the treatment group with a significant difference between groups favoring the treatment group. No significant difference in visual cognitions (measured by Benton Visual Retention Test) between groups. No significant difference in short-term memory (measured by S-PA) between groups.
	Vigna et al. (2019)[54]	Adults with overweight/obesity and at least one mood/sleep disorder or with binge eating behavior	Treatment (low-calorie diet + *H. erinaceus*): 35 Control (low-calorie diet): 35	8 weeks	1.2 g mycelium + 0.3 g fruiting body	No significant improvement in depression (measured by Zung's Depression Scale) from baseline in either group nor difference between groups, but in patients selected for depression symptomatology, only the treatment group showed significant improvements from baseline. Significant improvement in anxiety (measured by Zung's Anxiety Scale) from baseline in only the treatment group with even greater significance in patients selected for anxiety symptomatology in the treatment group but not control group. Significant improvement in "depression" and "anxiety" domains (measured by SCL-90) from baseline in the treatment group test with greater significance in patients selected for symptomatology.

	Li et al. (2020)[55]	Adults over 50 years old with Mild Alzheimer's Disease	Treatment: 20 Control: 21	49 weeks	1.05 g mycelia (enriched with 5 mg/g erinacine A)	No significant improvement in dementia-related neuropsychiatric symptoms (measured by NPI) in either group nor difference between groups. Trend of improvement in cognitive abilities (measured by CASI) from baseline in treatment group with a trend of deterioration in placebo group reaching significance at week 25, but no significant difference between groups. Significant improvement in cognitive function (measured by MMSE) from baseline in treatment group only, but no significant difference between groups. Significant difference in ability to perform instrumental activities of daily living (measured by IADL) between groups at the end of the intervention favoring the treatment group. No change in monocular or binocular BCVA (measured by Snellen eye chart) from baseline in either group. No change in monocular or binocular CS (measured by Pelli-Robson chart) from baseline in either group with the exception of a significant change in binocular CS in placebo from baseline and a significant difference in left-eye CS between groups favoring the treatment group.

AIS, Athens Insomia Scale; HAMD-17, 17-item Hamilton Depression Scale; POMS, Profile of Mood States; THI, Tinnitus Handicap Inventory; FACT-F, Functional Assessment of Cancer Therapy- Fatigue; EORTC QLQ-C30, European Organization for Research and Treatment of Cancer Core Quality of Life Questionnaire C30; HADS, The Hospital Anxiety and Depression Scale; ADAS-cog, Alzheimer's disease Assessment Scale-Cognitive; NPI, Neuropsychiatric Index; WHOQOL-BREF, World Health Org QoL questionnaire; FACT-G, Functional Assessment of Cancer Therapy - General; SHS, Subjective Happiness Scale; SWLS, Satisfaction with Life Scale; GDS, Geriatric Depression Scale; SF-12, Medical Outcomes Study Short Form-12 questionnaire; HDS-R, Revised Hasegawa Dementia Scale; CES-D, Center for Epidemiologic Studies Depression scale; ICI, Indefinite Complaints Index; MMSE, Mini-Mental State Examination; S-PA, Standard verbal paired-associate learning test; SCL-90, Symptom Checklist-90; CASI, Cognitive Abilities Screening Instrument; IADL, Instrumental Activities of Daily Living; BCVA, best-corrected visual acuity; CS, contrast sensitivity

[a]Number included in final analysis

[b]Part of a Reishi & Privet Formula containing 0.45 g dried sporederm-broken Ganoderma lucidum spores + 0.33 g glossy Privet fruit extract (3.36g total weight, standardized for >1% polysaccharide and > 2.0 mg oleanolic acid)

Coriolus versicolor (Trametes versicolor)

Changes in systemic oxidative stress, auditory function, and mood were measured in a population of patients with Meniere's disease (MD),[46] a chronic disorder of the ear, characterized by vertigo, tinnitus, and hearing loss related to oxidative stress[56] and subsequent cellular damage and cochleovestibular dysfunction.[57,58] In addition to significant favorable changes in markers of oxidative stress by *C. verisocolor*, significant improvements in tinnitus, measured by the Tinnitus Handicap Inventory (THI), and improvements in the areas of frequency range, average loss in dB, and intellection threshold related to auditory function were observed in the treatment group but not the control group. Notable improvements in the categories of "anger," "confusion," "depression," "fatigue," and "tension" in relation to MD from the Profile of Mood States (POMS) questionnaire were observed in the treatment group only. However, significance was not discussed.

Ganoderma lucidum

A significant improvement in "cognitive functioning" measured by the European Organisation for Research and Treatment of Cancer Core Quality of Life Questionnaire C30 (EORTC QLQ-C30) was observed in a population of patients with breast cancer following *G. lucidum* supplementation.[47] However, in a population of patients with Alzhiemer's disease (AD), *G. lucidum* supplementation did not significantly improve cognitive function compared to the control group measured by the Alzheimer's Disease Assessment Scale-Cognitive (ADAS-cog),[48] which includes assessment of memory, language, and praxis domains. Additionally, no significant improvement compared to the control group was observed in dementia-related symptoms measured by the Neuropsychiatric Index (NPI) in the same population with AD.

In females with fibromyalgia, happiness and satisfaction with life, measured by the Subjective Happiness Scale (SHS) and Satsisfaction with Life Scale (SWLS), improved significantly with *G. lucidum* intake, but not in the placebo group with a difference reaching significance between groups for happiness.[50] A significant improvement in depression measured by the Geriatric Depression Scale (GDS) was also observed in the *G. lucidum* group, which is consistent with the significant improvement in depression and anxiety, measured by the Hospital Anxiety and Depression Scale (HADS) in an earlier study on females with breast cancer.[47] However, despite both studies revealing significant improvements in the treatment groups but not control groups, a significant difference between groups was found only in the depression and anxiety of females with breast cancer[47] but not in the depression of females with fibromyalgia.[50] Furthermore, in the population of females with fibromyalgia, "mental health," measured by the Medical Outcomes Study Short Form-12 Questionnaire (SF-12), did not differ significantly between groups by the end of treatment.[50] There was also no significant difference in the "psychological" domain of quality of life measured by the World Health Organization Quality of Life (WHOQOL-BREF) questionnaire in patients with AD.[48]

Greater improvements in "emotional well-being" compared to controls were observed following *G. lucidum* intake in a population of adults with lung cancer[49] and with breast cancer[47] measured by Functional Assessment of Cancer Therapy: General (FACT-G) and Functional Assessment of CA Therapy: Fatigue (FACT-F) assessments respectively. "Emotional functioning" measured by the EORTC QLQ-C30 also improved significantly in the treatment group, but not control group, in females with breast cancer with a significant difference between groups.[47] In females with fibromyalgia, the "emotional role" of health, measured by the SF-12, improved significantly from baseline in the treatment group only, but the difference between groups was not significant.[50]

In patients with AD, the "social relationship" domain of quality of life, measured by the WHOQOL-BREF, did not improve significantly compared to the control.[48] Similarly, in females with breast cancer, *G. lucidum* did not result in any significant changes in "social/family well-being" or "social functioning," measured by FACT-F and EORTC QLQ-30 respectively.[47] Greater improvements in "social/family well-being" were observed in patients with lung cancer following intake of a *G. lucidum*-based formula compared to the control,[49] and significant improvements in "social functioning" were observed in females with fibromyalgia in the treatment group only.[50] However, neither study found significant differences between groups in these measures.

Hericium erinaceus

In a population of adults with mild cognitive impairment, cognitive function improved, measured by a modified Revised Hasegawa Dementia Scale (HDS-R), following *H. erinaceus* intake.[51] This change was significant from baseline and when compared to the placebo group. Similarly, in a population of adults with mild AD, significant improvements in cognitive function from baseline were observed following *H. erinaceus* intake measured by the Mini Mental State Examination (MMSE), which includes assessment of attention, orientation, memory, calculation, and language, though the difference was not significant when compared to the placebo.[55] MMSE scores also showed a significant improvement in a population of healthy adults over 50 years old following *H. erinaceus* intake with no change in the control group and a significant difference between groups.[53] However, although the change from baseline and difference between groups were statistically significant, the point-differences were miniscule with a change from baseline (treatment baseline: 29.19 vs. treatment end: 30.00) and difference between groups (treatment end: 30.00 vs. placebo end: 29.53) of less than one point out of total score of 30. Additionally, no significant change in short-term memory of words measured by the Standard Verbal Paired-associate Learning test (S-PA) was observed.

In the adults with mild AD, a trend of improvement in cognitive abilities was also observed in the treatment group with a corresponding decline in the placebo group which reached significance mid-way through the intervention.[55] This was measured by the Cognitive Abilities Screening Instrument (CASI), a tool used to screen for dementia which assesses attention, concentration, orientation, memory, language abilities, visual construction, list-generating fluency, abstraction, and judgment.[59] However, NPI scores in this population showed no significant improvement in frequency or magnitude of dementia-related neuropsychiatric symptoms which include delusions, hallucinations, dysphoria, anxiety, agitation and aggression, euphoria, disinhibition, irritability and lability, apathy, and aberrant motor activity.[60] Still, a significant difference in ability to perform instrumental activities of daily living (IADL), which indicates competence related to daily activities requiring thinking and organizational skills and a lower level of dependence, was observed between groups, favoring the treatment group. Also of note, though the difference did not reach significance, the treatment group showed greater retention of total neuronal fibers than the placebo group shown by magnetic resonance imaging (MRI).

In healthy adults over 50, visual cognitions measured by the Benton Visual Retention test, which includes measures of visual recognition, memorization, and construction, did not change following treatment.[53] In adults with mild AD, a significant difference was observed between groups in left-eye contrast sensitivity favoring the treatment group, but no other significant difference was found in any other measure of monocular or binocular visual acuity or contrast sensitivity measured by the Snellen eye chart and Pelli-Robson chart respectively.[55]

Trends of improvement in depression and anxiety were observed following *H. erinaceus* intake in females[52] and adults with overweight or obesity with at least one mood or sleep disorder or with binge eating behavior.[54] In adult females, depressive symptomatology, measured by the Center for Epidemiologic Studies Depression Scale (CES-D), improved significantly from baseline following *H. erinaceus* intake, but the difference was not significant between groups. Subscores of “incentive” and “concentration” relevant to depression and of “irritating” measured by the Indefinite Complaints Index (ICI) also improved significantly in the treatment group from baseline with a difference reaching significance for “incentive” compared to the placebo.[52] Similarly, in adults with overweight or obesity, depression, measured by Zung’s Depression Scale, improved significantly in treatment group patients selected for depression symptomatology. However, when not selected for symptomatology, there was no significant change in depression from baseline in either group nor significant difference between groups. Still, depression measured by the Symptom Checklist-90 (SCL-90) showed significant improvements in all treatment group participants with greater significance when selected for symptomatology. This was the same for the domain of anxiety measured by the SCL-90. Additionally, anxiety measured by Zung’s Anxiety Scale showed significant improvement from baseline with greater significance when selected for anxiety symptomatology. No significant change was seen in the control group.[54] Similarly, in adult females, a significant improvement in the “anxious” subscale of the ICI was shown following *H. erinaceus* intake. However, no significant difference was found between groups.[52]

DISCUSSION

The evidence of the beneficial impact of medicinal mushrooms on human cognition and mood is very limited with the lack of large human clinical trials with standardized procedures and methodologies. Still, from these studies, trends of improvement in cognition from *H. erinaceus* and in mood from *H. erinaceus* and *G. lucidum* appear. Following *H. erinaceus* intake, measures of cognitive function improved in adults with mild cognitive impairment[51] and with mild AD.[55] Measures of depression and anxiety improved significantly in adult females[52] and in adults with overweight or obesity with depression and anxiety symptomatology[54] following *H. erinaceus* intake. *G. lucidum* intake also appeared to produce significant improvements in emotional well-being or emotion-related health from baseline in adult females with fibromyalgia,[50] breast cancer,[47] and adults with lung cancer.[49] Additionally, improvements in emotional well-being, emotional functioning, and depression reached significance in a between-group comparison, favoring the treatment group, in the women with fibromyalgia.[50] Because the studies on *C. militaris* and *T. versicolor* were limited, with one human clinical trial each, no trends were identified nor the potential underlying mechanisms explored.

Nerve growth factor (NGF) is a type of neuropeptide in the neurotrophin family involved in the growth, maintenance, and protection of nerve cells. Brain-derived neurotrophic factor (BDNF), another neurotrophin, has a similar function in addition to its roles in facilitating neuroplasticity and memory. The reduced levels of neurotrophins present in neurodegenerative diseases and psychiatric disorders,[61,62] including Alzeihmer's and depression, point to the potential of neurotrophin-based therapies to address these conditions. The neuroprotective activity and cognition-enhancing effects of NGF in animal models of cognitive dysfunction[63–65] further support this potential for the treatment of AD. The efficacy of neurotrophin-based therapies can be understood through its protective and regenerative activity on cholinergic neurons demonstrated in animal models and in living patients which would counter the neuronal degeneration and loss leading to cognitive decline in AD.[66] Evidence for antidepressant-like effects by NGF administration in rats[67] also points to the potential for neurotrophin-based therapies in mood disorders. The "neurotrophic hypothesis of depression" directly relates decreased neurotrophin activity to depression with the neuronal atrophy, decreased neurogenesis, and loss of glia, and supports the key role of BDNF in antidepressant activity.[68,69] Neuroinflammation is also a prominent factor in the development and susceptibility to depression, with BDNF acting to mitigate neuroinflammation and high inflammation reducing the expression of BDNF.[70] While inflammation alone may not be enough cause for depression, its key role in the development or exacerbation of depression is supported by the reduced efficacy of conventional antidepressants in patients with high inflammation[70] and the greater efficacy of inflammation-reducing treatments, such as TNF-α inhibition[71] and antidepressants with secondary anti-inflammatory properties,[72] in depressed patients with high baseline inflammation.

H. erinaceus contains bioactive compounds shown to stimulate NGF-synthesis and subsequent neurite outgrowth *in vitro*.[73–77] Furthermore, *H. erinaceus* was shown to increase NGF levels in the locus coeruleus and hippocampi of rats[78] and to increase NGF mRNA expression in the hippocampi of mice,[73] suggesting the possibility that these compounds are capable of being absorbed into the blood and crossing the blood-brain barrier (BBB). These compounds include hericenones and erinacines among other unidentified bioactive compounds. In mice, oral administration of an *H. erinaceus* fruiting body powder resulted in an increased level of NGF mRNA expression in the hippocampus.[73] In this same study, Mori et al., found that the stimulation of NGF production by an ethanol extract of the *H. erinaceus* fruiting body in 1321N1 human astrocytoma cells appears to occur via activation of the c-Jun N-terminal kinase (JNK) signaling pathway, a mitogen-activated protein kinase (MAPK) signaling pathway involved in a variety of functions including cell proliferation, development and apoptosis. It is also a target for AD treatment due to its involvement in apoptosis regulation.[79] In another experiment, hericenone E appears to potentiate NGF-mediated neurogenesis via the MEK/ERK and PI3K-Akt signaling pathways in rat pheochromocytoma (PC12) cells.[77] These pathways are also involved in regulating normal cell proliferation, development, and survival. Erinacines also appear to act on NGF and BDNF by increasing the expression of these neurotrophins demonstrated *in vitro*[80] with further evidence of neuroprotective activity in aminal models of ischemic stroke[81] and Alzheimer's disease.[82] Antidepressant-like effects were also observed in several mouse models[83–85] with one study indicating antidepressant activity by the modulation of BDNF/PI3K/Akt/GSK-3β signaling by erinacine A-enriched mycelium[83] and another indicating antidepressant activity related to hippocampal neurogenesis.[84] Anti-inflammatory activity was also observed with decreased levels of IL-6 and TNF-α[83] and decreased TNF-α with an increase in IL-10 by amycenone,[85] another bioactive compound found in *H. erinaceus*. These compounds provide a solution to the limitations of direct neurotrophin administration such as the poor pharmacokinetic profile and inability of NGF to cross the BBB[86] as compounds that may cross the BBB[87,88] and stimulate neurotrophin synthesis.

Although the precise mechanisms are unknown, the outcomes of the studies included in this review are consistent with the theory that neurotrophin production stimulated by the bioactive compounds in *H. erinaceus* are protective against cognitive decline and may improve cognitive outcomes in the context of cognitive dysfunction. This is reflected by the improvements in cognitive function in populations with mild AD[55] and mild cognitive impairment[51] paired with the lack of improvement in a healthy population.[53] The mechanism behind effects of *H. erinaceus* on mood is also unclear, but may be related to the role of neurotrophins or other anti-inflammatory compounds in neuronal protection, regeneration, and the reduction of inflammation, which is implicated in pathophysiology of psychiatric disorders.

There is little information available on the potential mechanisms of action of *G. lucidum* on mood. In one study, *G. lucidum* demonstrates benzodiazepine-like activity via GABAergic mechanisms in rats.[89] This GABAergic activity may be related to the potential effects of *G. lucidum* on mood and anxiety as benzodiazepines, a class of drug used to treat anxiety and insomnia, work by facilitating the binding of GABA, a primary inhibitory neurotransmitter, to its receptors. *G. lucidum* triterpenoids also appear to act as neurotprohin mimetics, displaying neuronal survival-promoting activities[90] and inducing neuronal cell differentiation by MAPK activation[91] *in vitro*. Additional neuroprotective activity[92] and the down-regulation of mRNA expression of inflammatory mediators including tumor necrosis factor-α (TNF-α) and interleukin 1β (IL-1β) by *G. lucidum*[93,94] may also be related to its effects on mood through the mitigation of neuroinflammation, similar to the potential activity of *H. erinaceus*. However, more research is needed to pinpoint the precise mechanisms of action.

While the outcomes of the studies may appear very promising, study design must be taken into account and caution used when attempting to extend application. All but 2 of the 11 studies reviewed had less than 30 participants in the treatment group. 5 of the 11 studies reviewed had intervention durations of 2 months or longer. Several features of the studies also produce some ambiguity in the interpretation of the outcomes and conclusions. For one, the repeated administration of cognitive function tests may contribute to the learning effect which describes a pattern of an increasing score over time due to a familiarity with the test, skewing the results. For example, one study on adults with mild cognitive impairment supplementing with *H. erinaceus* showed gradual improvement in cognitive function scale scores in the placebo group in addition to the treatment group,[51] indicating that the seeming improvement may be due, at least in part, to the learning effect. Another consideration is the subjectivity of questionnaires. Affect and mood are difficult to quantify, and high variability between the reported results and the felt experiences of subjects are likely to be present due to the subjective nature of emotion.[95] The context of supplementation must also be considered as debilitating conditions such as cancer or fibromyalgia will impact mood as well. Improvements in one's physical condition may lead to improvements in mood, so it is difficult to isolate mood outcomes in disease states. In part, this is the nature of illness, but this also means that the studies on these sick populations do not have evidence to support claims, such as those of supplement companies, of generalized improvement in mood by medicinal mushrooms apart from these conditions.

The existing studies also vary greatly in the measurement tools used, duration of intervention, and form and dosage of mushroom used. Despite the common function of quantifying subjective measures such as "well-being" or "depression," the differences between the questionnaires will contribute to the variability between studies and study outcomes. The chosen components of mushrooms used in the studies also contribute to the variability despite the use of a common mushroom. These include the fruiting body, mycelium spores, or combinations of these. These components contain various types and amounts of bioactive compounds. For example, the hericenones of *H. erinaceus* are primarily found in the fruiting body and the erinacines in the mycelium,[87] so it is possible that the effect of *H. erinaceus* supplementation in a population of AD patients would differ if provided with a fruiting body-based supplement versus a mycelium-based supplement.

Dosages also varied between studies. The typical dosages based on the studies reviewed are 3 g for *G. lucidum* spore powder and 1-3.2 g for *H. erinaceus* fruiting body powder. The other forms and combinations were less frequently used and therefore do not have ranges. The amounts and forms are also detailed in Table 1. The dosing of *H. erinaceus* had a broader range, contributing to more variability. It is also notable that these studies demonstrated safety and a general tolerability of these mushrooms with few and minor side effects observed.

Future research may benefit from greater consistency in the measurement tools used between studies and consideration of frequency of cognitive function test administration to avoid the learning effect. Greater consistencies in the form, dose, and duration of supplementation would also reduce the variables between studies, allowing for more insightful comparisons and stronger evidence for use.

CONCLUSION

Modern research on functional mushrooms increasingly supports their therapeutic utility. This review reveals trends of improvement in well-being by *G. lucidum* in the context of taxing disease states, improvement in cognitive function by *H. erinaceus* in the context of cognitive decline, and general improvement in depression and anxiety by *H. erinaceus*. Medicinal mushrooms hold great potential as complements to limited conventional treatments. This is also supported by the low-risk and general safety of consumption. The evidence is still scarce with limitations in the size and number of studies, but this does not discount the potential efficacy. These studies show promising therapeutic application, general safety of mushroom consumption, and provide a foundation for the methods and procedures of future studies. This points to the need for larger human clinical trials to be conducted and further research to pinpoint the mechanisms behind the effects.

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